

Assessed Problem Sheet 2: A Crystal of Quantum Wells

1. Assessed Question. This problem sheet requires you to use the *Coupled Quantum Wells* Jupyter Notebook available on Blackboard. The notebook is available at...

PHYS60003/Course Content/Python Notebooks/

If the Notebook is not compatible with your python installation, then please use the “Manual” version, also available at the address above.

The Jupyter notebook introduces the computational method, but this is NOT required knowledge. Energy eigenvalue and Wave-function solutions can be studied by using the parameter control sliders in Figure 1, or via the code cell below the heading “Set Parameters Manually” for the manual version. In this problem sheet, you will explore how these solutions map onto the tight binding model solutions, which we introduced in lectures. For the enthusiasts among you, the notebook also describes how to make your own calculations. To complete the assessed parts, numerical answers must be uploaded on Blackboard via the APS 2 test in the folder **PHYS60003/Course Content/Problem Sheets/Assessed Problem Sheets/**

- (i) **Computing 1D quantum well states.** [*Not assessed.*] Set the simulation tool to compute the energy spectrum of a single 1D quantum well with parameters: well width, $w = 2.5$ nm; barrier height, $V = 0.3$ eV; and barrier width, $b = 0.75$ nm.
 - (a) Check the simulator gives you quantum well states with energies: $E_0 = 0.036$ eV, $E_1 = 0.140$ eV and $E_2 = 0.302$ eV. Energies can be output via the code cell below the Figure 1.
 - (b) Verify that the probability density and amplitude of each quantum well state are as you would expect for the first 3 states. To toggle amplitude and probability plots, click on the red buttons next to the peak energy selector.
 - (c) Please feel free to adjust parameters to see how they affect the energy spectrum and eigenvalues.
 - (d) Return the simulation tool to the original quantum well parameters specified above before continuing with the tasks below.
- (ii) **Mapping 1D quantum well array states to a tight binding band.** Adjust the number of coupled quantum wells to $N = 21$. The crystal period is $a = w + b = 3.25$ nm. You should find a series of bands in the energy spectrum; within a band, each peak is a quantum state of the finite chain of quantum wells. Check that there are 21 states in each band; can you explain this observation? We will use α as a band label; it indicates the quantum well states from which a band is formed. The lowest energy band for $\alpha = 0$ is called Band 0, etc. To check that your simulator is set up correctly, the middle (11^{th}) state of Band 0 should have an energy of $\approx E_0 = 0.036$ eV; can you explain this observation?

The energy spectrum produced by the simulation tool can be mapped onto the tight binding model, where the energy for the bands can be written as,

$$E_{\alpha k_n} = E_\alpha - 2t_\alpha \cos(k_n a),$$

where E_α is the energy of state α of a single quantum well, t_α is the hopping energy for Band α and k_n are a discrete wavenumbers of states identified by the simulation tool. Answer the following questions and enter your answers on Blackboard. (Please check that your simulation is set up correctly using the values for E_α given above. As a double check, the lowest value energy state should be $\min\{E_{\alpha k}\} = 0.032$ eV.)

(a) Estimate the hopping energy t_0 from the width of Band 0. [1 mark]

(b) What is the sign of the hopping energy t_0 ? [1 mark]

[Hint: consider the phase between neighbouring quantum wells in the wavefunction at the top and bottom of the band.]

(c) Estimate the relative e^- effective mass (m^*/m_0) at the bottom of Band 0. [2 marks]

(d) Estimate the hopping energy t_1 from the width of Band 1. [1 mark]

(e) What is the sign of the hopping energy t_1 ? [1 mark]

(f) Estimate the relative e^- effective mass (m^*/m_0) at the bottom of Band 1. [2 marks]

(iii) **Bloch wavefunctions, boundary conditions and k-states.** The Bloch wavefunctions for the finite 1D crystal may be written as,

$$\psi_{\alpha k_n}(x) \approx A \sin(k_n x) \phi_{\alpha k_n}(x)$$

where $\phi_{\alpha k_n}(x + na) = \phi_{\alpha k_n}(x)$ is the periodic lattice function. This assumes boundary conditions, $\psi_{\alpha k_n}(L) = \psi_{\alpha k_n}(0) = 0$ where, $L = Na + b \approx Na$, is the crystal length. (Ignore the last barrier for simplicity.)

(a) What values of k_n do the lowest energy states of Bands 0 and 1 take? *[Note: Use the boundary conditions above and not those used in lectures.]* [2 marks]

(b) *Optional:* Try to directly fit the tight bonding model to the peak energies, $E_{\alpha k_n}$, from the python model and the discrete values of k_n implied from part (iii)(a). (“print(selectPeak.options)” outputs peak energies.) The python data do not fit the model so well as $\psi_{\alpha k}(x)$ “leaks” out of the crystal. Can you find a better fit of model and data when L is a free parameter?

[Total 10 marks]